

Machine Learning Lab – Question 5

Question

Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42, stratify=iris.target)

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
pred = knn.predict(X_test)

print("K =", 5)
print("Accuracy:", round(accuracy_score(y_test, pred), 4))
```

Output

```
K = 5
Accuracy: 0.9667
```